TOSHKENT SHAHRIDAGI INHA UNIVERSITETI
INHA UNIVERSITY IN TASHKENT

STUDENT NAME:	
STUDENT ID NUMBER:	
TION NUMBER IN THIS COURSE:	

HW1 – FALL 2025

COURSE NAME:	*Big Data Analytics*
COURSE NUMBER:	SOC4170
EXAMINATION DATE:	
TIME:	
EXAMINATION DURATION:	
ADDITIONAL MATERIALS ALLOWED TO USE:	**Calculators are allowed**
SPECIAL INSTRUCTIONS:	*- Answers will only be evaluated if they are readable.* *- Show all your work. For some questions, you may get partial credit even if the end result is wrong due to a calculation mistake. If you make assumptions, state your assumptions clearly and precisely.*

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST:

Desks should be free from all unnecessary items (books, notes, technology, food, water, clothes)

Use of any electronic device (Phone, iPod, iPad, laptop) is not allowed during the examination.

Cheating, talking to fellow students, singing, turning back are not allowed

Write your Name (*capital letters*), ID number and Section number in each page of your examination paper.

Final answers must be written by **only blue or black, non-erasable pen**. Do not use highlighters or correction pen.

All answers should be written in the space provided for each question, unless specified the other way.

If you have a problem please raise your hand and wait quietly for a Proctor.

You are not allowed to leave the exam room until you submit the exam papers.

STUDENT NAME:

STUDENT ID NUMBER: TION NUMBER IN THIS COURSE:

1.[62 points] Frequent Pattern Mining & Word Count

We have 4 transactions as follows;

T1 = {A, B, C, D}

T2 = {A, C, D, E}

T3 = {A, B, C, E}

T4 = {A, B, C, E}

The minimum support is 70% and the minimum confidence is 80%.

(a) [4 Points] Let 4 Mappers M1 – M4 take transaction data T1 – T4 respectively, write the output (list of <key, value> pairs) of each mapper to come up with frequent 1-itemsets.

Output of M1 = < A, 1 >, <B, 1>, <C, 1>, <D, 1>
Output of M2 = < A, 1 >, <C, 1>, <D, 1>, <E, 1>
Output of M3 = < A, 1 >, <B, 1>, <C, 1>, <E, 1>
Output of M4 = < A, 1 >, <B, 1>, <C, 1>, <E, 1>

(b) [11 Points] Let 5 Reducers R1 – R5 receive <key, (value1, value2, …)> pairs, write the input of each reducer, write the output <key, value> pair of each reducer, and write frequent 1-itemsets.

Input of R1 = < A, (1, 1, 1, 1) >
Input of R2 = < B, (1, 1, 1) >
Input of R3 = < C, (1, 1, 1, 1) >
Input of R4 = < D, (1, 1) >
Input of R5 = < E, (1, 1, 1) >

Output of R1 = < A, 4 >
Output of R2 = < B, 3 >
Output of R3 = < C, 4 >
Output of R4 = < D, 2 >
Output of R5 = < E, 3 >

List of frequent 1-itemsets: {A, B, C, E}

(c) [2 Points] Write 2-itemsets of self-join result. Also write candidate 2-itemsets.

2-itemsets of self-join result: {A, B}, {A, C}, {A, E}, {B, C}, {B, E}, {C, E}

candidate 2-itemsets: {A, B}, {A, C}, {A, E}, {B, C}, {B, E}, {C, E}

STUDENT NAME:			
STUDENT ID NUMBER:		TION NUMBER IN THIS COURSE:	

(d) [4 Points] Let 4 Mappers M1 – M4 take transaction data T1 – T4 respectively, write the output (list of <key, value> pairs) of each mapper to come up with frequent 2-itemsets.

Output of M1 = <A:B, 1>, <A:C, 1>, <B:C, 1>
Output of M2 = <A:C, 1>, <A:E, 1>, <C:E, 1>
Output of M3 = <A:B, 1>, <A:C, 1>, <A:E, 1>, <B:C, 1>, <B:E, 1>, <C:E, 1>
Output of M4 = <A:B, 1>, <A:C, 1>, <A:E, 1>, <B:C, 1>, <B:E, 1>, <C:E, 1>

(e) [13 Points] Let 6 Reducers R1 – R6 receive <key, (value1, value2, …)> pairs, write the input of each reducer, write the output <key, value> pair of each reducer, and write frequent 2-itemsets.

Input of R1 = <A:B, (1, 1, 1)>
Input of R2 = <A:C, (1, 1, 1, 1)>
Input of R3 = <A:E, (1, 1, 1)>
Input of R4 = <B:C, (1, 1, 1)>
Input of R5 = <B:E, (1, 1)>
Input of R6 = <C:E, (1, 1, 1)>

Output of R1 = <A:B, 3>
Output of R2 = <A:C, 4>
Output of R3 = <A:E, 3>
Output of R4 = <B:C, 3>
Output of R5 = <B:E, 2>
Output of R6 = <C:E, 3>

List of frequent 2-itemsets: {A, B}, {A, C}, {A, E}, {B, C}, {C, E}

(f) [4 Points] There are 2 machines Ma1 and Ma2. Both machines contain the same list of frequent 2-itemsets. Assign index number(s) to Ma1 and Ma2 so that they carry out the same number of combinations. For example, if Ma1 contains index 1, it generates the combination of the first 2-itemset with the rest of all the 2-itemsets.

Index number(s) assigned to Ma1: 1, 4

Index number(s) assigned to Ma2: 2, 3

STUDENT NAME:

STUDENT ID NUMBER: ꟷTION NUMBER IN THIS COURSE:

(g) [4 Points] Write the output of the mapper M1 running on Ma1 and the output of the mapper M2 running on Ma2

Output of M1 = <A:B:C, 1>, <A:B:E, 1>, <A:B:C, 1>, <B:C:E, 1>

Output of M2 = <A:C:E, 1>, <A:B:C, 1>, <A:C:E, 1>, <A:C:E, 1>

(h) [4 Points] Write the output of the combiner C1 running on Ma1 and the output of the combiner C2 running on Ma2

Output of C1 = <A:B:C, 1>, <A:B:E, 1>, <B:C:E, 1>

Output of C2 = <A:C:E, 1>, <A:B:C, 1>,

(i) [4 Points] Write the candidate 3-itemsets and the frequent 3-itemsets

candidate 3-itemsets: {A, B, C}, {A, C, E}

frequent 3-itemsets: {A, B, C}, {A, C, E}

(j) [12 Points] Write all the rules with support and confidence.
Rule1: { } -> { }, xx%, yy%
Rule2: { } -> { }, xx%, yy%
...
where xx% is support, and yy% is confidence.

STUDENT NAME:

STUDENT ID NUMBER: CTION NUMBER IN THIS COURSE:

Rule1: {B} -> {A} 75% 100%
Rule2: {A} -> {C} 100% 100%
Rule3: {C} -> {A} 100% 100%
Rule4: {E} -> {A} 75% 100%
Rule5: {B} -> {C} 75% 100%
Rule6: {E} -> {C} 75% 100%
Rule7: {A, B} -> {C} 75% 100%
Rule8: {B, C} -> {A} 75% 100%
Rule9: {A, E} -> {C} 75% 100%
Rule10: {C, E} -> {A} 75% 100%
Rule11: {B} -> {A, C} 75% 100%
Rule12: {E} -> {A, C} 75% 100%

2.[17 points] Page Rank

We have 4 documents as follows;

D0: BTS BLACK PINK MUSIC AWARD

D1: BLACK PINK COLOR

D2: SONG BLACK PINK

D3: MUSIC BTS

D0 has hyperlinks to D1, D3

D1 has a hyperlink to D2, D3

D2 has a hyperlink to D0, D1

D3 has a hyperlink to D0

(a) [4 Points] The equation to get the page rank of a document is as follows;

$$r_i = d \left[\sum_{p_j \in \mathcal{I}(p_i)} \frac{r_j}{c_j} \right] + (1-d)\frac{1}{|D|}$$

Where d = 0.8. And d is multiplied only by reducers.
Each document's initial page rank is 10
Let 4 Mappers M0 - M3 take documents D0 – D3 respectively, write the output (list of <key, value> pairs) of each mapper to come up with the page rank of each document.
Round off the number to the nearest tenth. For example, 6.65 is 6.7 and 3.33 is 3.3.

Output of M0 = <D1, 5>, <D3, 5>
Output of M1 = <D2, 5>, <D3, 5>
Output of M2 = <D0, 5>, <D1, 5>
Output of M3 = <D0, 10>

STUDENT NAME:

STUDENT ID NUMBER: TION NUMBER IN THIS COURSE:

(b) [4 Points] We have 4 Reducers R0 – R3. Write the output of each reducer to come up with page rank of each document. For simplicity, **ignore** the second part of the equation.
Round off the number to the nearest tenth. For example, 6.65 is 6.7 and 3.33 is 3.3.

Output of R0 = <D0, 12>
Output of R1 = <D1, 8>
Output of R2 = <D2, 4>
Output of R3 = <D3, 8>

(c) [8 Points] Assume that we need one more cycle of Map/Reduce to come up with the page rank of each document with the same number of mappers and reducers. Write the output of mappers and reducers. Again, for simplicity, **ignore** the second part of the equation.
Round off the number to the nearest tenth. For example, 6.65 is 6.7 and 3.33 is 3.3.

Output of M0 = <D1, 6>, <D3, 6>
Output of M1 = <D2, 4>, <D3, 4>
Output of M2 = <D0, 2>, <D1, 2>
Output of M3 = <D0, 8>

Output of R0 = <D0, 8>
Output of R1 = <D1, 6.4>
Output of R2 = <D2, 3.2>
Output of R3 = <D3, 8>

(d) [1 Points] Now, we have the page rank of each document. Write the sequence of documents when a user typed BLACK PINK in the search window and clicked "search" button.

D0 -> D1 -> D2

STUDENT NAME:

STUDENT ID NUMBER: ... TION NUMBER IN THIS COURSE:

3.[11 points] Decision Tree

The table below represents a training set, D, of class-labeled tuples. The class label "buys_computer" has two distinct values (yes, no). The attribute list contains "age", "income", "student", "credit_rating".

RID	Age	Income	Student	Credit_rating	Buys_Computer
1	Youth	High	Yes	Fair	Yes
2	Youth	Medium	No	Excellent	Yes
3	Youth	High	Yes	Fair	Yes
4	Middle-aged	High	Yes	Excellent	Yes
5	Middle-aged	High	No	Fair	No
6	Middle-aged	Medium	No	Fair	Yes
7	Middle-aged	Medium	No	Excellent	No
8	Senior	High	Yes	Fair	No
9	Senior	Low	No	Excellent	No
10	Senior	Low	No	Excellent	No

And the table below represents $\log_2 n$ **from n=1 to n=18**

N	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\log_2 n$	0	1	1.6	2	2.3	2.6	2.8	3	3.2	3.3	3.5	3.6	3.7	3.8	3.9	4	4.1	4.2

$$Info(D) = -\sum_{i=1}^{m} p_i \log_2(p_i) \qquad Info_A(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} \times Info(D_j) \qquad Gain(A) = Info(D) - Info_A(D)$$

(a) [2 Points] What is Info(D)? Show all your work.

$-(\frac{1}{2}\lg_2\frac{1}{2} + \frac{1}{2}\lg_2\frac{1}{2}) = 1$

(b) [4 points] What is $Info_{Age}(D)$ and $Info_{Income}(D)$? Show all your work.

$Info_{age}(D) = 0.3 \times (-(1 \lg_2 1)) + 0.4 \times (-(\frac{1}{2}\lg_2\frac{1}{2} + \frac{1}{2}\lg_2\frac{1}{2}))$
$+ 0.3 \times (-(1 \lg_2 1)) = 0.4$

$Info_{income}(D) = 0.5 \times (-(\frac{3}{5}\lg_2\frac{3}{5} + \frac{2}{5}\lg_2\frac{2}{5})) +$
$0.3 \times (-(\frac{2}{3}\lg_2\frac{2}{3} + \frac{1}{3}\lg_2\frac{1}{3})) +$

$0.2 \times (-(1\lg_2 1)) = 0.75$

STUDENT NAME:

STUDENT ID NUMBER: ~~SE~~CTION NUMBER IN THIS COURSE:

(c) [4 Points] What is $Gain_{age}(D)$ and $Gain_{income}(D)$? Show all your work.

$Gain_{age}(D) = 0.6$

$Gain_{income}(D) = 0.25$

(d) [1 Points] Which attribute can be the splitting criterion between age and income?

Age

3.[10 points] Confusion Matrix

We have 2 models, M1 and M2 which detect fraud credit card transactions.

Total number of test data transactions are 1000. And the test data contains 4 actual fraudulent transactions.

(a) [5 Points] M1 detected 3 fraudulent transactions, one of which is a real fraudulent transaction. Complete the confusion matrix and compute accuracy, precision, and recall.

Predicted class / Actual class	Fraud = Yes	Fraud = No	Total
Fraud = Yes	1	3	4
Fraud = No	2	994	996
Total	3	997	1000

Accuracy: 99.5 % (1 points)

Precision: 33.3 % (1 points)

Recall: 25 % (1 points)

STUDENT NAME:

STUDENT ID NUMBER: ⌐TION NUMBER IN THIS COURSE:

(b) [5 Points] M2 detected 10 fraud transactions, 4 of which are fraud transactions. Complete the confusion matrix and compute accuracy, precision, and recall.

Predicted class Actual class	Fraud = Yes	Fraud = No	Total
Fraud = Yes	4	0	4
Fraud = No	6	990	996
Total	10	990	1000

Accuracy: 99.4 % (1 points)

Precision: 40 % (1 points)

Recall: 100 % (1 points)